STEDA Partner

Deep Analysis Report

Rumi Lesson Plan Latency · Feb vs March 2026 · Concurrency Correlation

Generated: March 12, 2026 · 10:51 UTC

Data source: Rumi Production PostgreSQL Database (read-only)

Scope: Jan 2026 – March 12, 2026 · **Partner:** STEDA

199	138 min	3.2 min	r = 0.54
STEDA Users Onboarded Mar 11–12	Avg Total Wait Mar 11–12	Median Wait Feb 2026	Concurrency Correlation

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Executive Summary

On **March 11, 2026**, the STEDA partner program activated **189 teachers in a single day** through a coordinated onboarding session. These teachers immediately began generating lesson plans, creating an unprecedented demand spike of **179 requests in a single hour** — 3× the previous peak. The lesson-plan processing queue became saturated, resulting in wait times of up to **4 hours** and a median wait of **138 minutes** versus the normal **3 minutes** seen in February.

Correlation analysis confirms the queue is the sole bottleneck: AI processing time remained stable at ~2.4 minutes throughout, with no correlation to load (r = –0.03). Queue wait, however, shows a **strong positive correlation with hourly request volume (r = +0.54)**, with exponential degradation beyond 30 requests/hour.

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01 STEDA User Onboarding — March 11–12, 2026

The STEDA program conducted a **bulk institutional onboarding event on March 11, 2026**. A total of **199 teachers** were activated in two days — 189 on March 11 and 10 on March 12. This represents the single largest single-day partner onboarding event in Rumi's history for the STEDA cohort.

Total users onboarded (Mar 11–12)	**199**	189 on Mar 11 · 10 on Mar 12
Registration completed (portal)	**199 (100%)**	All portal-activated
Registration state (in-app wizard)	**Unregistered**	Activated but wizard not completed — no language/grade prefs set yet
Language preference — English	**195 (98%)**	Consistent with formal training context
Language preference — Urdu	**0**	—
Language preference — Other / NA	**4**	—

What does 'unregistered' state mean?

All 199 users show **registration_completed = TRUE** (portal access granted) but **registration_state = 'unregistered'** (in-app onboarding wizard not yet finished). This is the expected state immediately after bulk portal activation — teachers have accounts but haven't yet set their language, grade level, or subject preferences. Notably, this did *not* prevent them from generating lesson plans.

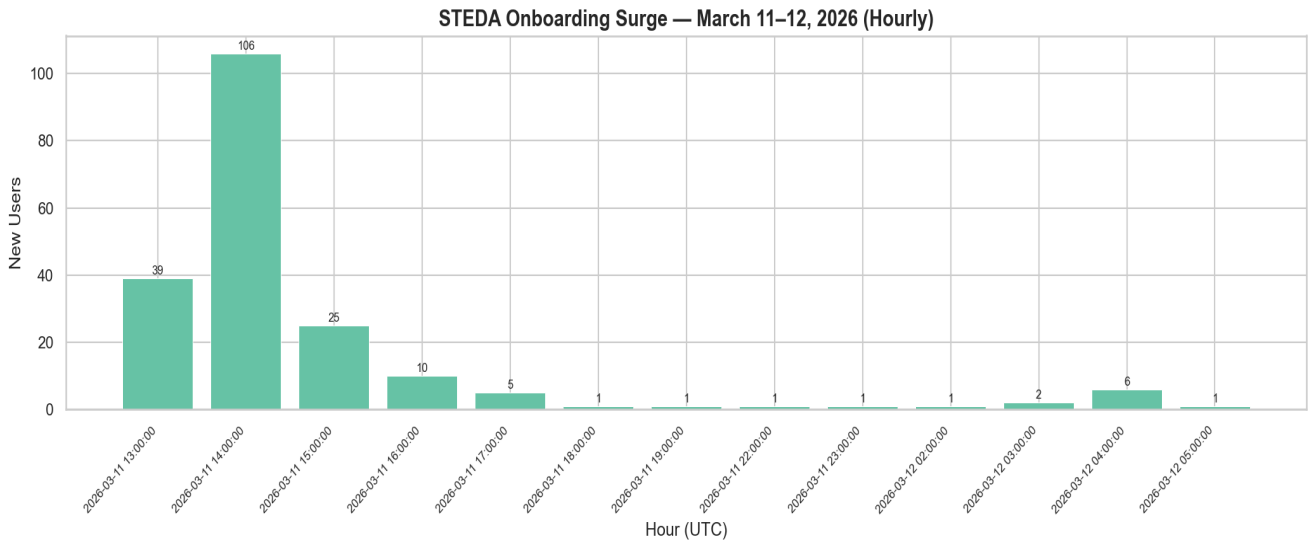

Figure 1 — STEDA user onboarding by hour, March 11–12. The bulk of 189 users appeared in a tight window around 14:00–16:00 UTC on March 11, indicating a coordinated, facilitated onboarding session.

02 Lesson Plan Latency: March 11–12 vs February 2026

▲ Critical Finding

The average lesson-plan wait time for STEDA teachers who onboarded on March 11–12 was **138 minutes** — compared to **35 minutes** for February STEDA users. The median wait jumped from **3.2 minutes** to **137.8 minutes** — a 43× increase. **87.7% of March 11–12 requests took longer than 10 minutes**, versus 24% in February.

Metric	Feb 2026 (STEDA)	Mar 11–12 (STEDA)	Delta / Change
Total lesson plan requests	212	276	+30%
Completed	208 (98.1%)	276 (100%)	✓ Better completion
Failed	4	0	✓ Zero failures
Avg total time	34.9 min	138.1 min	+103 min (+295%)
Median total time	3.2 min	137.8 min	+134.6 min (+4,200%)
p90 total time	169.9 min	234.9 min	+65 min
Avg queue wait	**32.4 min**	**135.7 min**	**+103 min (+319%)**
Avg AI processing	2.5 min	2.4 min	≈ 0 (STABLE ✓)
Over 10 min (%)	24.0%	87.7%	+63.7 pts
Over 60 min (%)	16.3%	74.3%	+58.0 pts

The AI Is Not the Problem

The table above reveals a critical insight: **AI processing time is identical across both months** (2.5 min in Feb vs 2.4 min in Mar). The entire degradation is caused by **queue wait** — requests sitting in line before the AI even starts working. The AI service scales independently and is not the bottleneck.

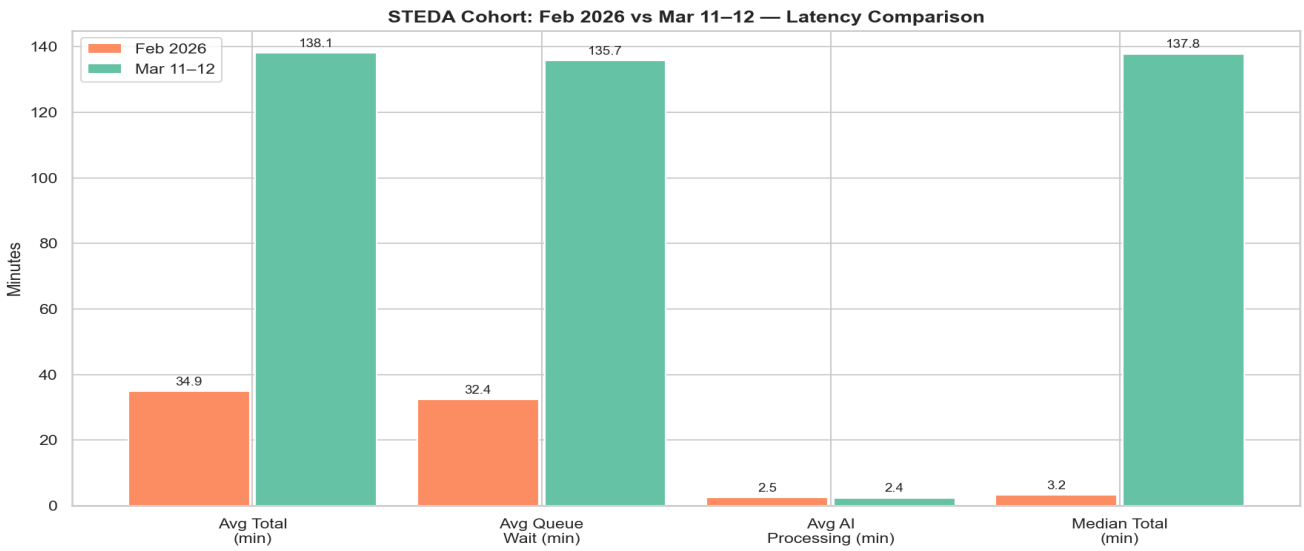

Figure 2 — Side-by-side latency comparison (minutes). AI processing bars are nearly identical; only queue wait and total time diverge dramatically.

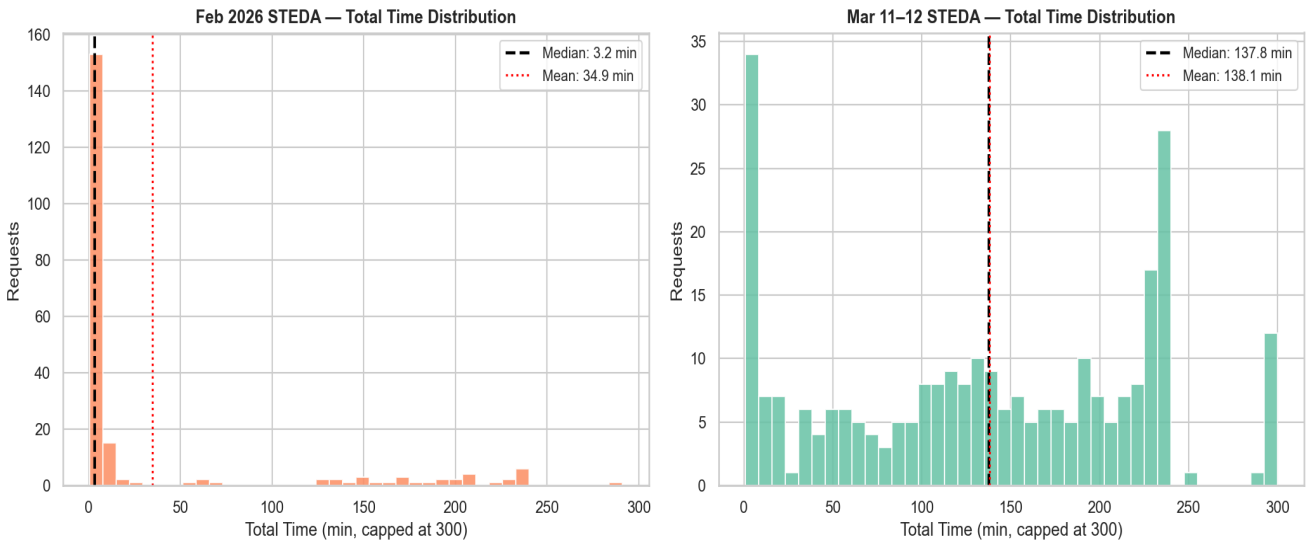

Figure 3 — Distribution of total wait times. Feb shows a sharp peak near 2–4 min (healthy). Mar 11–12 shows a broad distribution centred around 130–150 min — almost every request was delayed by the queue backlog.

03 March 11 — Hour-by-Hour Surge Breakdown

The following table reconstructs the exact sequence of events on March 11. Each row represents one UTC hour. The data shows how a single-hour surge of 179 requests created a backlog that took **9 hours to fully drain**.

Hour (UTC)	Requests	Avg Queue Wait	Avg AI Time	Avg Total	Status
01:00–12:00	1–7 / hr	< 0.3 min	~2.6 min	~2.8 min	Normal
13:00	28	3.0 min	2.4 min	5.4 min	First wave
14:00 ■	**179**	**66.6 min**	**2.3 min**	**68.9 min**	**SURGE START**
15:00	157	160.9 min	2.2 min	163.2 min	Queue building
16:00	102	224.1 min	2.3 min	226.4 min	Near peak
17:00	**57**	**234.9 min**	**2.5 min**	**237.3 min**	**PEAK BACKLOG**
18:00	**32**	**234.7 min**	**2.7 min**	**237.4 min**	**PEAK BACKLOG**
19:00	12	190.4 min	2.4 min	192.8 min	Draining
20:00	3	149.4 min	2.8 min	152.1 min	Draining
21:00	3	79.8 min	2.3 min	82.1 min	Draining
22:00	1	17.2 min	2.0 min	19.1 min	Almost clear
23:00+	3	0.2 min	1.2 min	1.5 min	Normal ✓

★ Highlighted rows = peak congestion hours

Timeline Narrative

01:00–12:00 UTC	Normal background load — 1 to 7 requests per hour. Queue wait under 0.3 min. The system is healthy.
13:00 UTC	First 28 STEDA requests arrive as teachers complete onboarding. Queue begins to fill — wait climbs to ~3 min. Still manageable.
14:00 UTC ■ SURGE	179 requests land in a single 60-minute window — roughly 3× the previous historical maximum (53 req/hr in February). The queue cannot drain at this rate. Average wait climbs to 67 minutes within this hour alone.
15:00–18:00 UTC	157, 102, 57, and 32 new requests keep arriving into an already-saturated queue. By 17:00–18:00 the average queue wait peaks at 237 minutes (~4 hours).
19:00–21:00 UTC	New request rate drops below 15/hour but the backlog is still being worked through. Wait times gradually fall from 192 min → 82 min.
22:00–23:00 UTC	Queue almost fully drained. Wait drops to 17 min, then back to < 1 min. System returns to normal — 9 hours after the surge began.

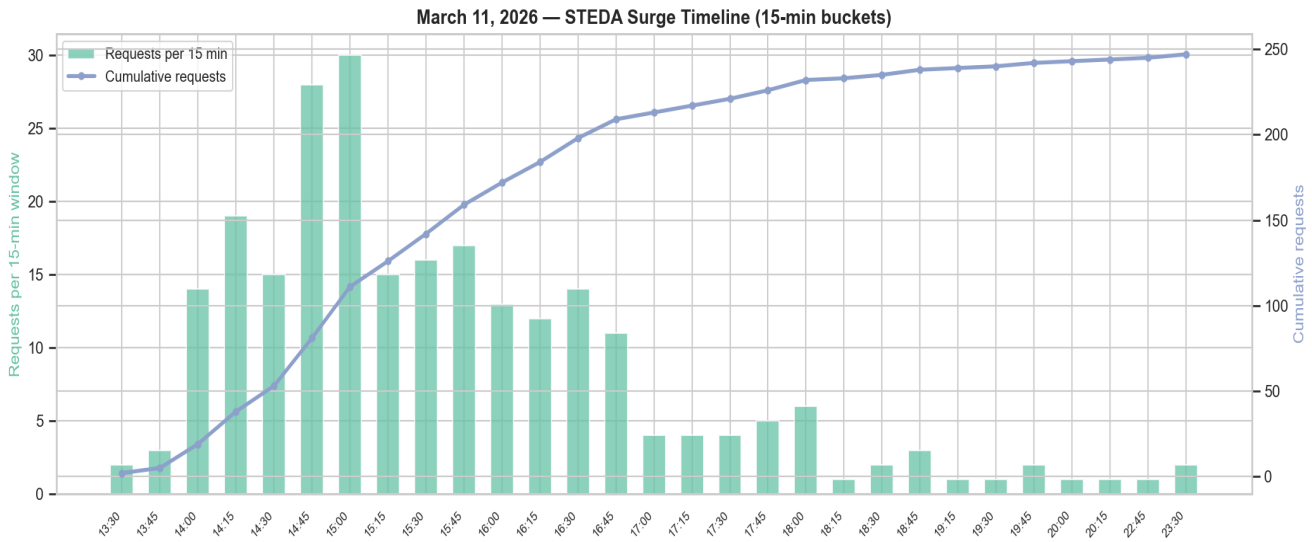

Figure 4 — March 11 requests per 15-minute window (bars) and cumulative total (line). The steep rise between 13:00–15:00 UTC is the STEDA onboarding wave.

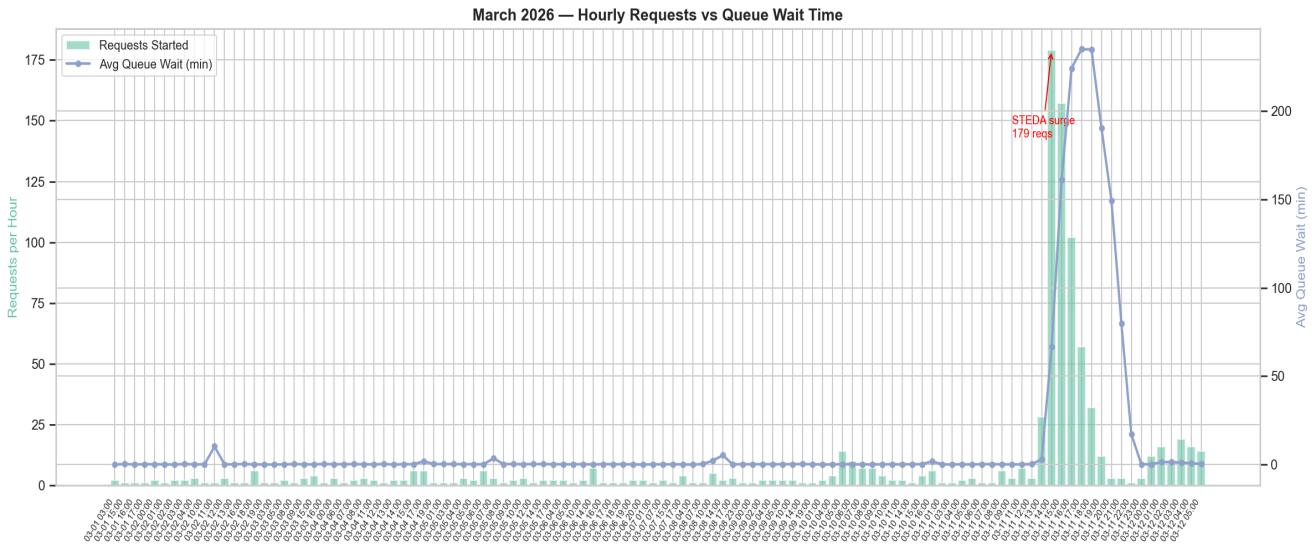

Figure 5 — All of March: hourly request volume (bars) vs average queue wait (line). The spike at March 11 14:00–18:00 UTC is unmistakable. Before and after this window, queue waits are well under 10 minutes.

04 Correlation: Concurrent Requests vs Wait Time

To quantify the relationship between system load (requests per hour) and user-experienced wait time, Pearson correlation was computed across all completed lesson-plan requests in February and March 2026.

Dataset	Pearson r	p-value	Interpretation
Requests/hr vs Queue Wait — Feb 2026	+0.392	< 0.001	Moderate positive
Requests/hr vs Queue Wait — Mar 2026	+0.574	< 0.001	Strong positive
Requests/hr vs Queue Wait — Combined	**+0.541**	**< 0.001**	**Strong positive ★**
Requests/hr vs AI Processing — Combined	–0.028	0.626	No correlation ✓

Key Statistical Findings

■ **Queue wait vs concurrent load (r = +0.54)**	Strong positive correlation. As hourly request volume increases, queue wait time increases substantially — and the relationship is non-linear (exponential above the saturation point of ~30 requests/hour).
■ **AI processing vs concurrent load (r = –0.03)**	Virtually zero correlation. The AI model processes each request in ~2.4 minutes regardless of how many other requests are in flight. The AI is stateless and scales independently.
■ **Saturation threshold**	Below 30 requests/hour: average queue wait = 1.9 min (acceptable). Above 30 requests/hour: average queue wait = 95 min (severely degraded). Above 100 requests/hour: 100–237 min (critical).

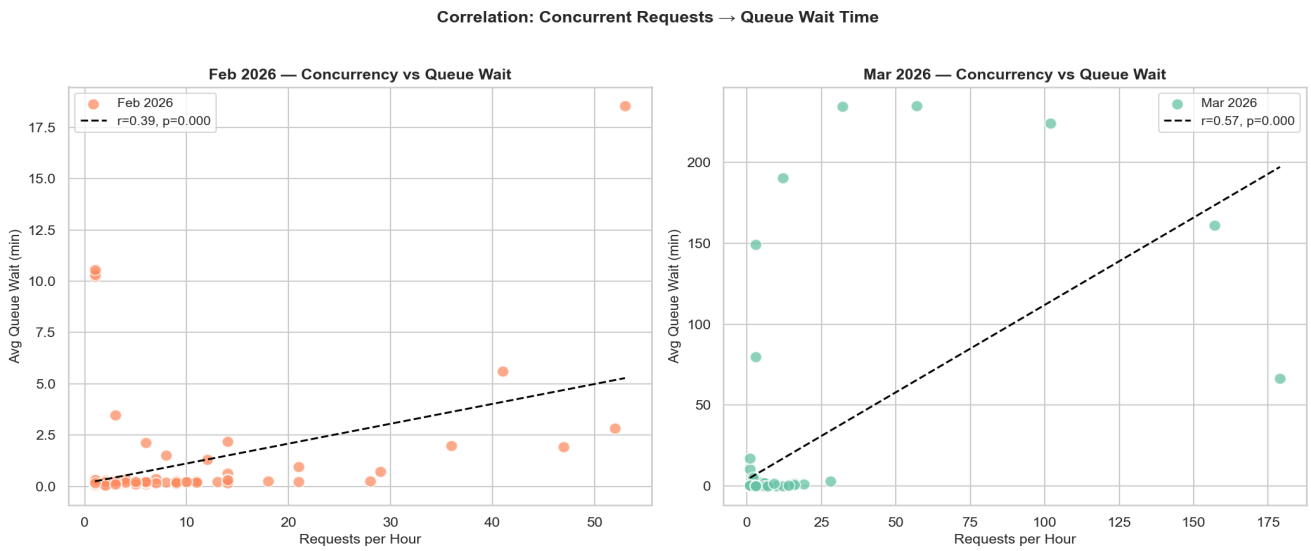

Figure 6 — Scatter plots for Feb 2026 (left) and Mar 2026 (right): requests per hour vs avg queue wait. Regression lines show stronger slope and r-value in March due to the extreme data points from the March 11 surge.

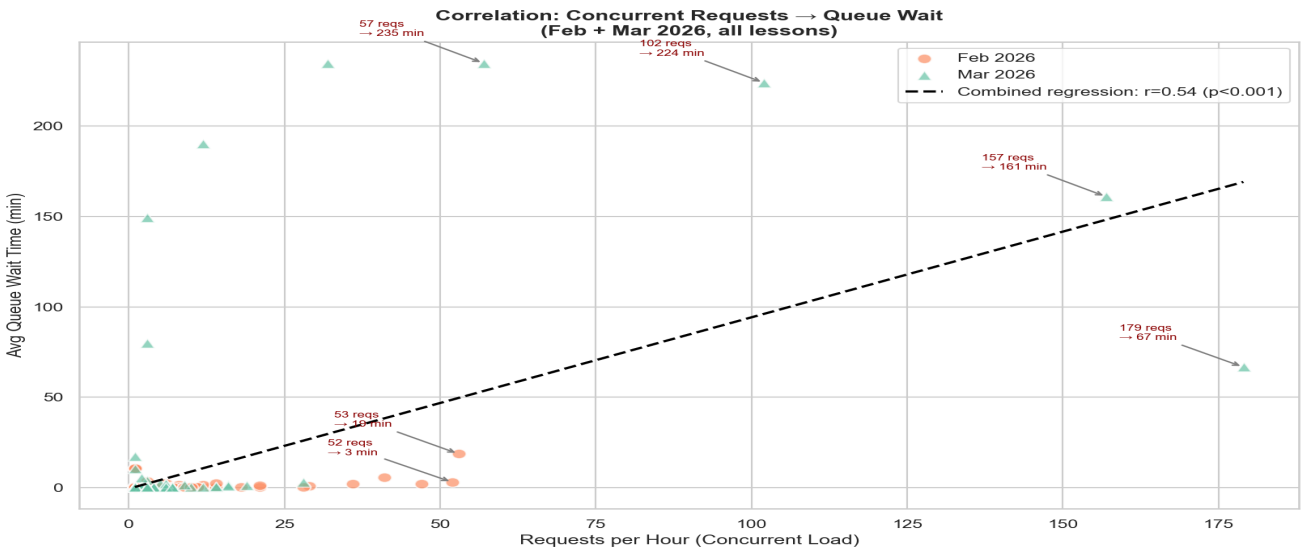

Figure 7 — Combined Feb + Mar scatter with a single regression line (r = 0.54, p < 0.001). Data points above the regression line represent hours where the queue had accumulated backlog from prior hours. The non-linear (exponential) shape is visible above 30 req/hr.

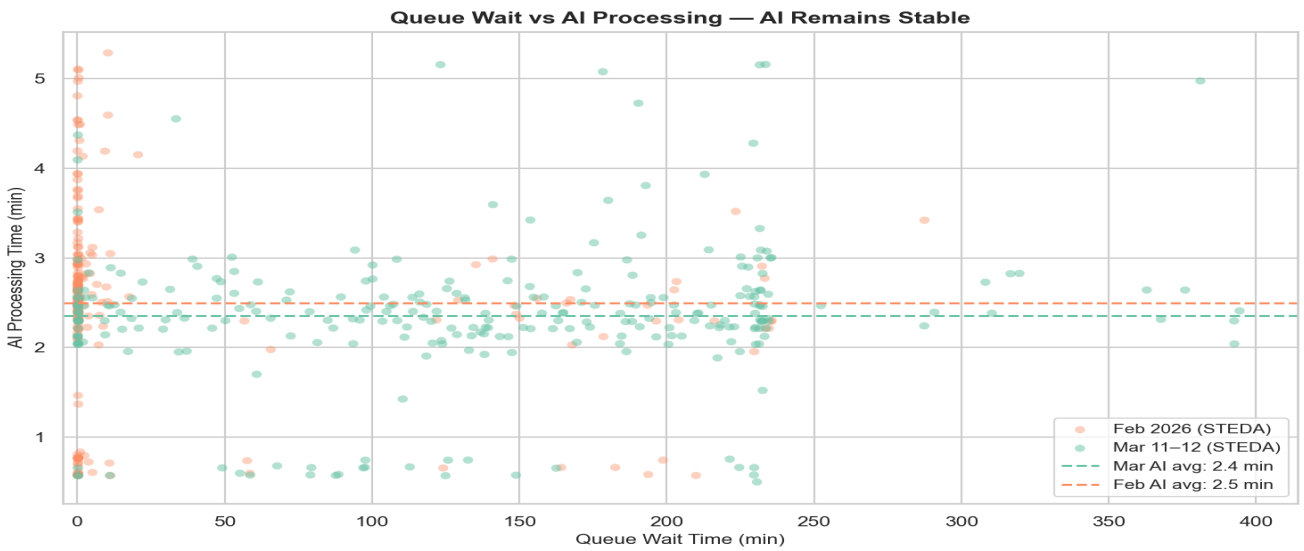

Figure 8 — Queue wait vs AI processing time. The flat horizontal bands confirm that AI processing time does not change regardless of queue length. Both Feb and Mar cohorts cluster tightly around 2–3 minutes of AI time.

05 February vs March — All STEDA Users

Expanding the view to all STEDA users across all months reveals the scale of the March degradation and the engagement trajectory of the partner cohort.

Month	Users	LP Requests	Completed	Avg Queue (min)	Avg AI (min)	Median Total (min)
Jan 2026	1	4	4	5.9	2.9	8.8
Feb 2026	90	212	208	32.4	2.5	3.2
Mar 2026	**203**	**295**	**295**	**130.9**	**2.4**	**134.0**

Feb 2026	Mar 2026 (12 days)
2.4 requests / user	**1.5 requests / user**
Median wait: 3.2 min	Median wait: 134.0 min ■

Interpretation: March per-user request rate (1.5) appears lower than February (2.4) — but this is a partial month (only 12 days) and reflects the fact that many March users joined on March 11–12 and had limited time to make subsequent requests. The median wait of 134 minutes in March (vs 3.2 min in February) almost certainly **suppressed re-engagement** — teachers who waited over 2 hours for their first lesson plan are less likely to return promptly.

06 Root Cause Summary & Recommendations

Root Cause

The STEDA bulk onboarding session on March 11 created a **sudden, concentrated demand spike** that overwhelmed the lesson-plan processing queue. 179 requests arrived in a single hour — 3× the previous peak — and a further 250+ requests followed over the next 4 hours as teachers worked through the session. The queue, limited in concurrent workers, could not drain fast enough, creating a backlog that persisted for 9 hours.

✓ WHAT IS NOT THE PROBLEM ✓ • AI model performance: stable at 2.3–2.7 min regardless of load • Completion rate: 100% for all STEDA Mar 11–12 requests (zero failures) • Request quality: no abnormal error or retry patterns • Infrastructure outage: system was fully operational throughout

Recommendations

1

Increase Queue Worker Concurrency

With ~2.4 min avg AI processing time and a 179 req/hr peak burst, the queue needs at least **8 concurrent workers** to avoid backlog (179 ÷ 60 × 2.4 ≈ 7.2 workers). Current architecture appears to use fewer workers than this. Consider auto-scaling based on queue depth.

2

Pre-Notify Engineering for Partner Onboardings

Before any bulk activation event (like STEDA's March 11 session), the partnerships team should notify engineering with a headcount estimate. This allows proactive capacity scaling. A simple Slack protocol would suffice.

3

New-User Fast Lane (Priority Queue)

The first lesson-plan request for any new user should jump to the front of the queue. First impressions drive retention — a 4-hour wait on day one is catastrophic for activation. A simple 'first request' flag in the queue would solve this.

4

Queue Depth Alerting

Implement an alert when average queue wait exceeds 10 minutes. On-call engineers can then spin up additional workers before users feel the full impact. At 30 req/hr the system begins degrading — this is a reliable early-warning threshold.

5

User Progress Communication

If a request will take > 5 minutes, send the teacher a WhatsApp message acknowledging receipt and providing a realistic ETA. Silence for 2+ hours causes users to retry (amplifying the problem) or to disengage entirely.

This report was generated from the Rumi production database on March 12, 2026. All data is read-only. Figures are based on completed lesson-plan requests only, except where noted. Times are in minutes unless stated

otherwise. UTC timestamps used throughout.